Marketing Analytics

Prediction, Validation, and Marketing Actions

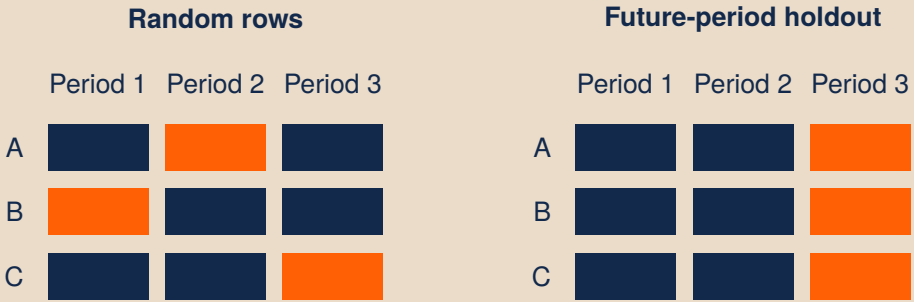
By Nathan Yang

Session 11

A Service-Capacity Decision

- A team can review 100 customers each week.
- Predictions use information available before that week.
- The outcome and review benefit determine the useful metric.

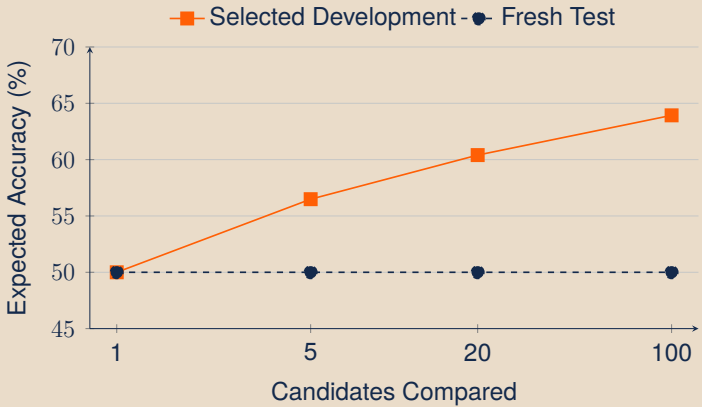
Validation Reproduces Deployment



Blue: training. Orange: test. Rows identify customers.
For new-customer deployment, hold out complete customer rows instead.

The deployment population determines the appropriate holdout.

Selection on Development Accuracy



Synthetic independent chance predictions: 80 development cases per candidate.

The Forecast Origin

- Features end before the decision time.
- The outcome window begins after the decision.
- Training labels must already be observable at fitting time.

Baselines and Model Families

- Compare with a constant probability or recent-history forecast.
- Logistic regression supplies a probability baseline.
- Trees represent conditional splits. Ensembles combine trees.

Logistic Prediction

Transform a linear index into a binary-outcome probability.

$$\widehat{p}_i = \frac{1}{1 + \exp[-(\beta_0 + x_i^\top \beta)]}$$

A coefficient changes log odds. Its probability effect depends on the starting point.

Preprocessing Belongs in Training

- Fit imputation, scaling, and vocabulary on training records.
- Choose model settings using validation data.
- Keep the final test sample separate from model selection.

Rare Purchases and Accuracy

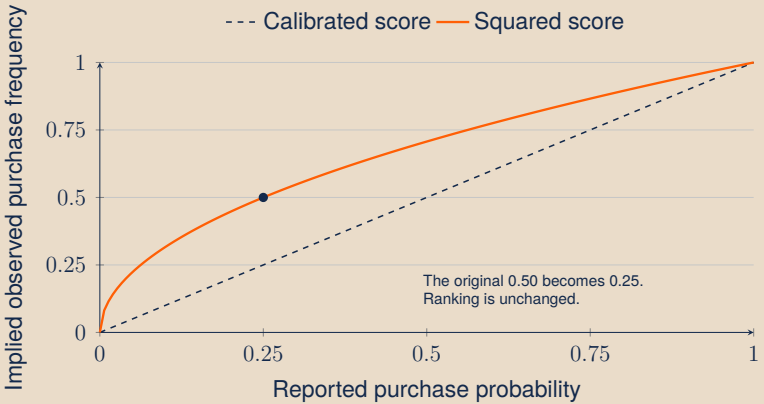
Quantity	Flagged	Not Flagged
Purchases	30	20
No Purchase	70	880

Synthetic population: 1,000 customers and 50 purchases.

Metrics and the Review Queue

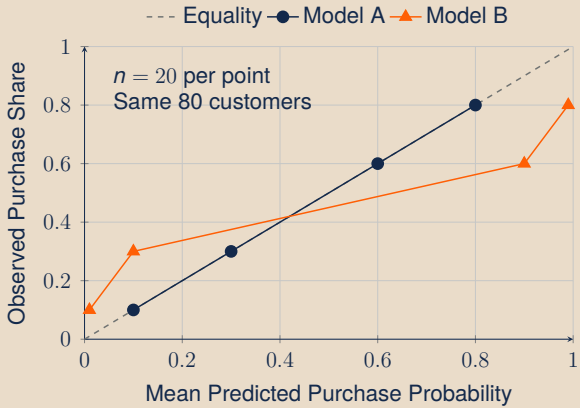
- Precision: what fraction of reviewed cases are positive?
- Recall: what fraction of positives enter the queue?
- Calibration: do predicted probabilities match observed frequencies?

Ranking Can Stay Fixed While Probabilities Change



A synthetic transformation squares initially calibrated probabilities.

Identical Ranking, Different Probabilities



Synthetic customers and imposed scores: both models have AUC 0.803. Each plotted group contains 20 customers.

AUC Alone Does Not Set Inventory

Quantity	Model A	Model B
AUC	0.803	0.803
Brier Score	0.175	0.21855
Predicted Purchases	36	40
Observed Purchases	36	36

Same 80 synthetic customers. Smaller Brier scores indicate less squared probability error.

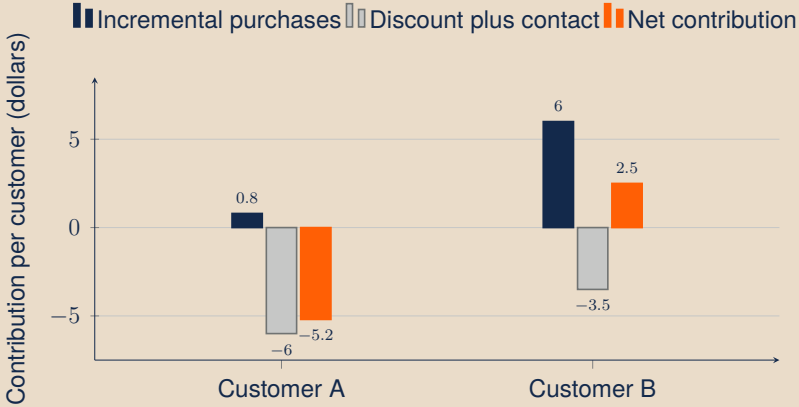
A Cost-Based Threshold

Suppose false positives cost C_{FP} and false negatives cost C_{FN}.

$$p > \frac{C_{FP}}{C_{FP} + C_{FN}}$$

With costs 1 and 9, the threshold is 0.10 under this simplified classification loss.

Coupon Targeting Includes the Untreated Outcome



Synthetic probabilities and costs. High treated response can have negative incremental value.

Queue Metrics

Compute metrics from the synthetic confusion matrix.

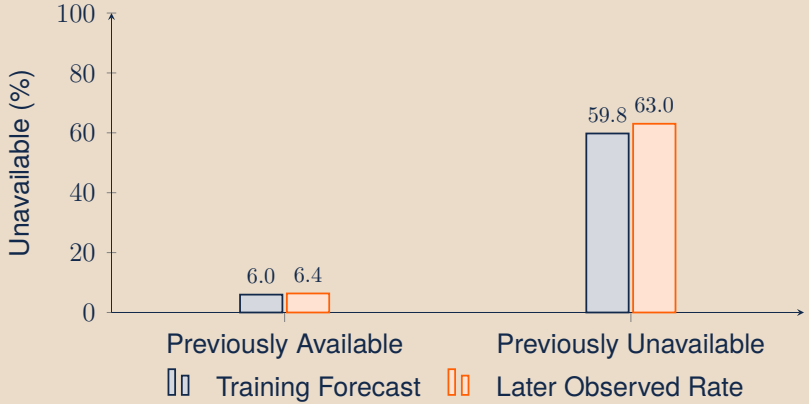
```
tp, fn, fp, tn = 30, 20, 70, 880
precision = tp / (tp + fp)
recall = tp / (tp + fn)
accuracy = (tp + tn) / (tp + fn + fp + tn)
assert (precision, recall, accuracy) == (0.3, 0.6, 0.91)
```

The model finds 60% of purchasers while 70% of its queue contains nonpurchasers.

McBroken Forecasting

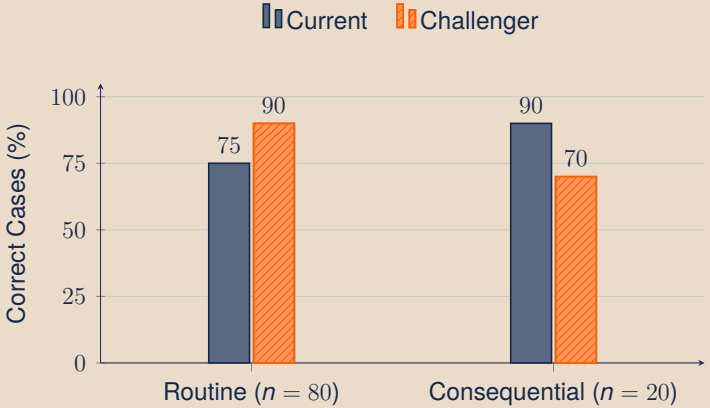
- Predict availability on a later date.
- Retain a separate count of unknown future labels.
- Compare baseline and fitted forecasts on the same observed outcomes.

Prediction and Outcome Coverage



McBroken forecasts use later dates and observed-label denominators.

Accuracy and Failure Costs



Constructed cases: pooled accuracy rises from 78% to 86%.

Paired Regression Checks

The same one hundred constructed cases are scored under both configurations.

```
improved, regressed, cases = 14, 6, 100
gain_pp = 100 * (improved - regressed) / cases
old_loss = 20 * 1 + 2 * 20
new_loss = 8 * 1 + 6 * 20
print(gain_pp, old_loss, new_loss)
```

Output: 8.0 percentage points; current loss 60 dollars; challenger loss 128 dollars.

Coding Exercise: A Forecast Evaluation

- Compare development and fresh accuracy after candidate selection.
- Check every feature against the forecast date.
- Compare a simple baseline, ranking, and calibration.
- Report missing outcomes beside the test metrics.

Targeting and Delivered Opportunities

- An apparently neutral target can produce unequal delivery.
- Evaluate exposure and error costs across relevant groups.
- Housing, credit, and employment applications require their specific legal context.

Prediction Discussion

- Which error uses scarce review capacity?
- Which feature becomes available too late?
- Does the score support a forecast, an intervention, or both?